



14-763/18-763

Systems and Toolchains for AI Engineers

FALL 2026

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Agenda

- Welcome and Introductions
- The Data Science Process
- Expectations for Incoming Students
- Teaching Team Introductions
- Course Syllabus & Schedule
- Course Dataset: NSL-KDD
- Next Steps

What is this course about?!

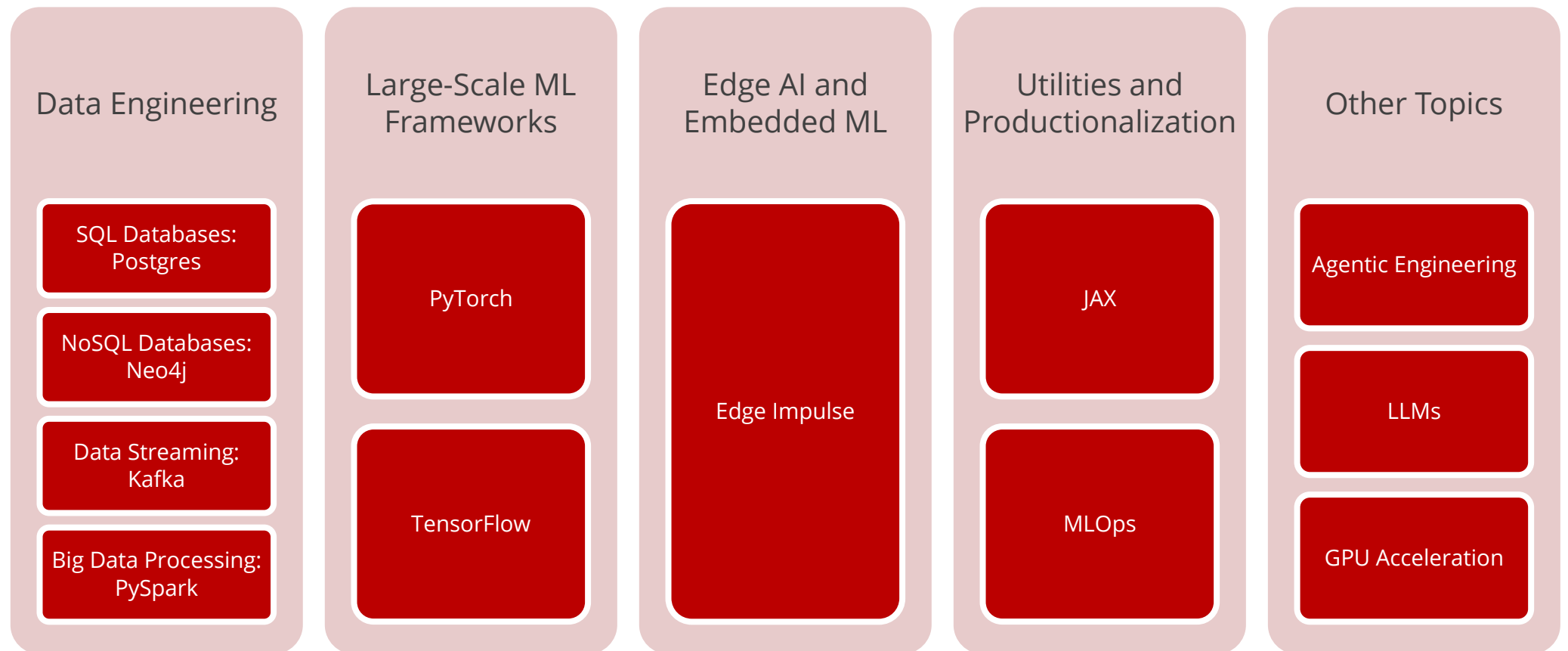


Figure 1.0 Course Technologies

Course Knowledge Depth

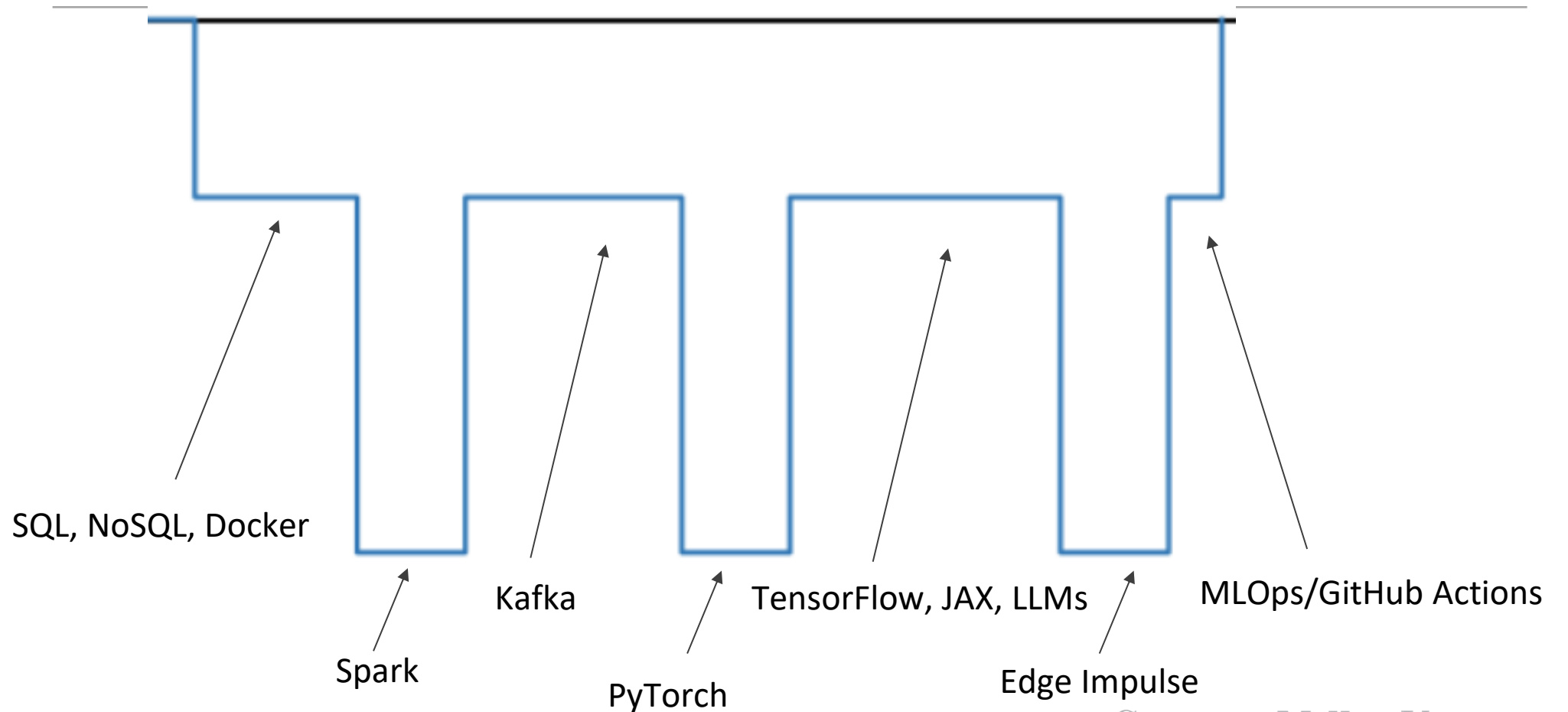


Figure 2.0 Course Knowledge Depth

Relevance of Course Topics to Today's Market

Machine learning is not only about models.

ML Systems and Toolchains aim to productionalize models

AI Engineering and Generative AI are areas of focus in this course

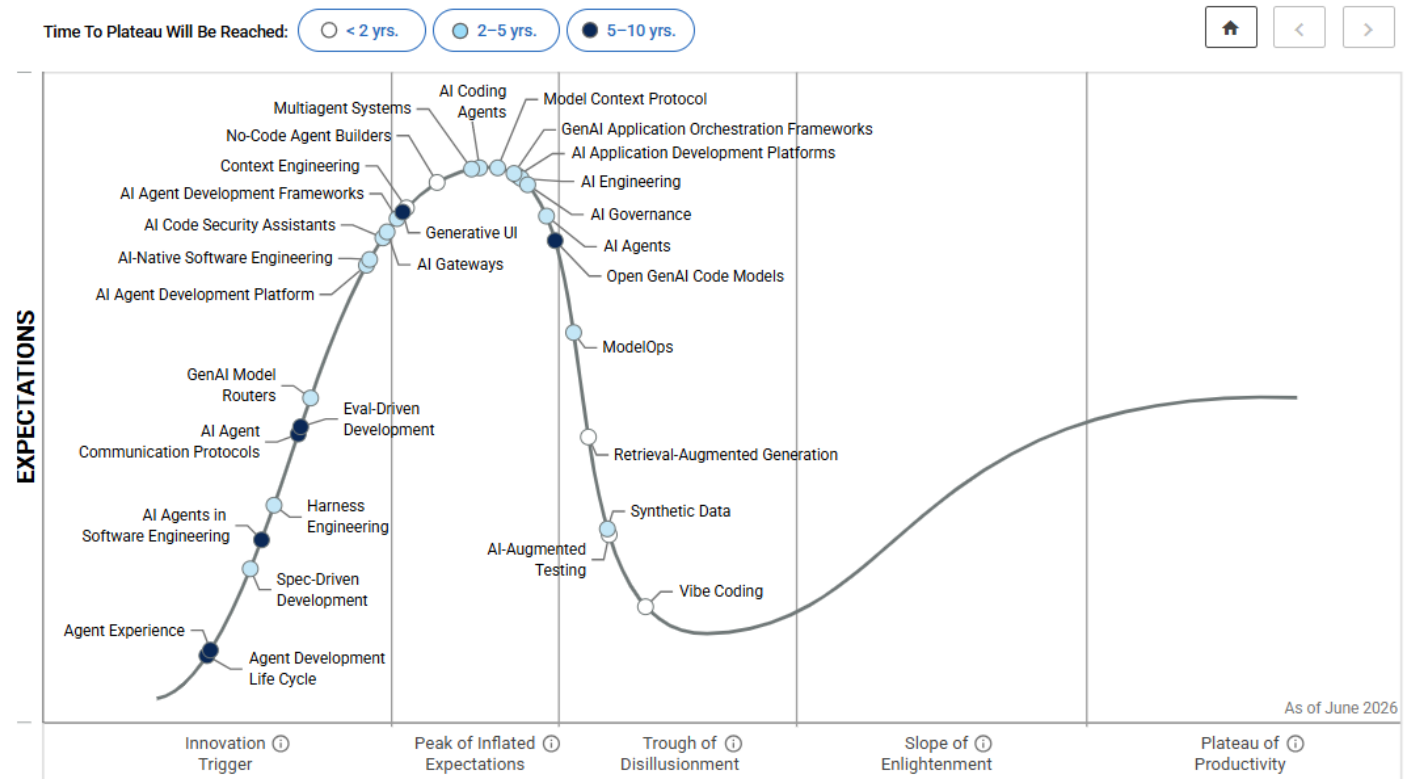


Figure 3.0 2026 Hype Cycle for AI in Software Engineering, by Gartner



Why is this course Important?

- This course provides you with the foundational practical knowledge on the most prominent AI tools and frameworks in the industry.
- An ad hoc approach to AI isn't sustainable and won't fulfill the expected market growth & demand.
- The use of advanced AI tools, frameworks and practices will enable scaling and operationalizing AI, leading to sustainable AI that meets our market demand.
- [PwC analyzed job growth from 2024 to 2025](#) and found AI specialist job postings soared (68.9% rise) while total job growth rose only 8.6%
- The Machine Learning market has a relentless pace of innovation, reflected by multiple trends such as democratization, augmentation, operationalization and composability. This innovation is reflected in growth opportunities and increase market size.
 - “AI Market value of nearly 100 billion U.S. dollars is expected to grow twentyfold by 2030, up to nearly two trillion U.S. dollars.”, Statista Research.

The Machine Learning Modeling Workflow

Machine Learning requires a lot more than just training!

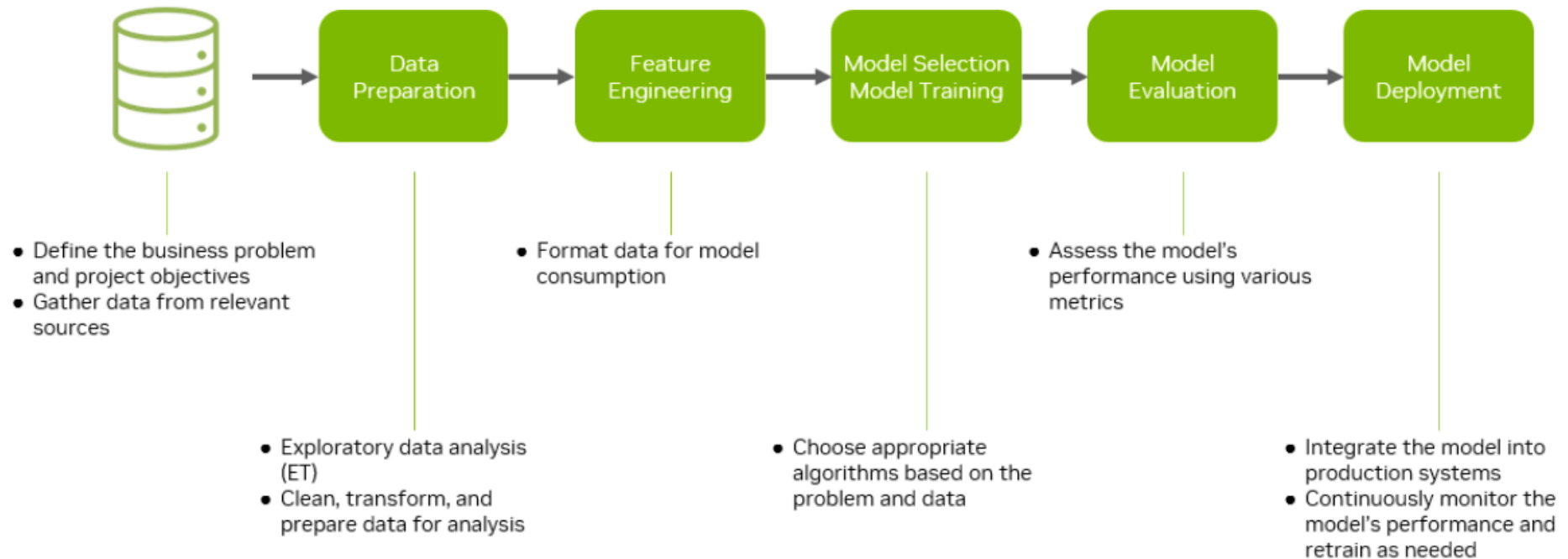


Figure 4.0 The Machine Learning Modeling Workflow



The Machine Learning Modeling Workflow – Cont'd

- Data may be sourced from IoT devices, logs from web servers, data gathered from social media, census datasets, data streamed from online sources using APIs, etc.
- Data preparation, or cleaning, is the process of collecting, choosing, modeling, and transforming data to answer an analytical question.
- Feature engineering is an iterative process that aims to transform raw data into AI-Ready format that is easier for algorithms to learn patterns.
- Model selection and training is the process of building the algorithms which learn to make predictions about unforeseen/future data.
 - The efficiency & accuracy of the machine learning model is then **evaluated** and fine-tuned to provide best possible performance.
- Model deployment refers to the application of a machine learning model on new data in a production environment. This process includes monitoring the model and communicating the model results to the users and data scientists



What this course is and What is it not?

- This course will help you get an introduction on different tools and technologies that are needed to build machine learning models at large-scale or for embedded systems.
- This course will help you productionalize your models by running them on the cloud and as part of a larger pipeline using MLOps
- This course is **NOT a source of learning the internals of machine learning models.**
- Also, this course doesn't offer a lot of depth on the theory behind the design of machine learning models.



Expectations for Incoming Students

- *You are expected to know Python or are willing to learn it.*
- *You are expected to have an introductory knowledge about machine learning models and a basic understanding of neural network theory.*

Instructor Introductions

MOHAMED FARAG

FARAG@CMU.EDU

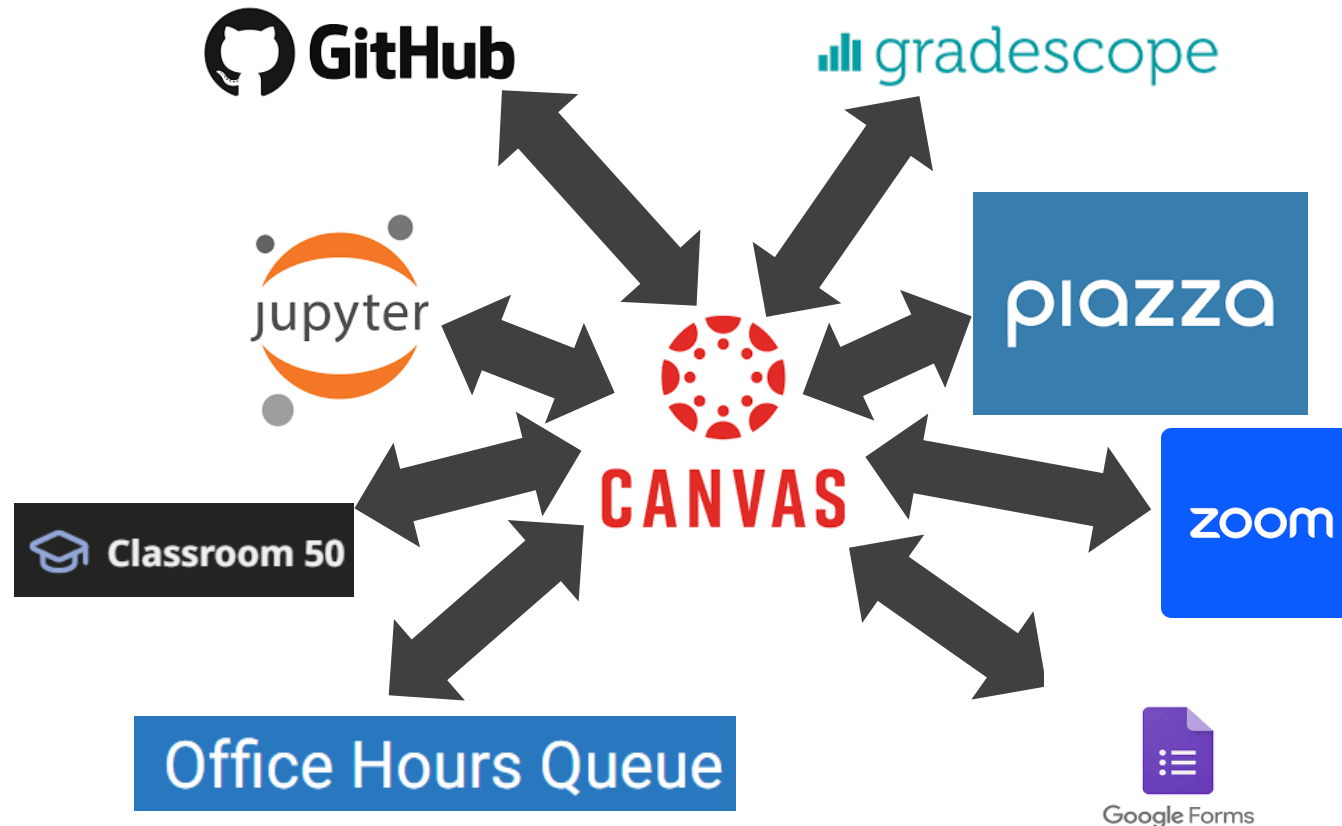
GUANNAN QU

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TA Introductions

- Diego Barrientos Pereira: dbarrien@andrew.cmu.edu
- Jiawei (Riva) Sun: jiaweis2@andrew.cmu.edu
- Megha Khelgi: mkhelgi@andrew.cmu.edu
- Myra Cropper: mcropper@andrew.cmu.edu
- Pranav Bhosale: pbhosale@andrew.cmu.edu
- Prarthana Rout: prout@andrew.cmu.edu
- Serena Gomez: serenag@andrew.cmu.edu
- Shashank Aluru: yaluru@andrew.cmu.edu
- Tiffany Bao: tbao2@andrew.cmu.edu

Course Delivery Technology Stack



Course Logistics



- Lectures recordings are made available after the lecture end time.
- Future lectures will be posted on Canvas as Jupyter Notebooks and PDF files.
- You may ask questions interactively or via Google Form during the lecture.
Google Form Link is: <https://forms.gle/JbKyKYAUGNrMs4oa9> or use the above QR code.
- Use the Student Space Slack Channel to find a teammate for your course project (No instructor or TA help is offered there)
- Students who have approved accommodation shall contact the course instructor to figure out how the instructor can meet their needs

Course Logistics – Office Hours (Zoom)

In-person/Zoom OHs												
Days/Timeframes	8-9am ET	9-10am ET	10-11am ET	11am-12pm ET	12:00-1:00pm ET	1-2pm ET	2-3pm ET	3-4pm ET	4-5pm ET	5-6pm ET	6-7pm ET	8-9pm ET
Monday			Pranav	Riva	Mohamed/Guannan							Prathana
Tuesday			Diego	Riva/Pranav			Shashank			Serena		
Wednesday		Megha					Shashank				Pranav	
Thursday	Serena			Myra/Diego		Megha	Shashank	Guannan	Mohamed	Prathana	Tiffany	Riva
Friday				Prathana/Myra								
Saturday			Serena									
				Instructor Office Hours - Conducted remotely via Zoom - URL can be found on Canvas								
				Instructor Office Hours - Starting week-7								
				TA Office Hours - Conducted remotely via Zoom - URL can be found on Canvas								
				In-Person OHs. Location: CIC Building. Room: 1313								
				10-min Slots by appointment only. Booking Link can be found on Canvas								

- Instructor Office Hours, Recitations and TA Office Hours URLs can be found on Canvas.
- For urgent questions, you can book a 10-minute meeting on Thursday 6-7pm.
 - Spots are limited! Don't overuse this feature.
 - Link to schedule the meeting can be found on Canvas.

Course Logistics – Piazza Hours

Piazza OHs								
	9-10am ET	10-11am ET	1-2pm ET	3-4pm ET	4-5pm ET	5-6pm ET	7-8pm ET	10-11pm ET
Monday	Tiffany					Riva		
Tuesday	Megha				Serena		Riva	
Wednesday		Diego	Pranav					Shashank
Thursday	Myra			Tiffany			Prathana	Shashank
Friday	Tiffany							
Saturday		Megha			Pranav			
Sunday		Prathana						

- Use Course Piazza to ask asynchronous questions that require instructor and/or TA help
- Please note that TAs will respond to inquiries/questions made ***before*** the Piazza OHs start time. Questions and inquiries that are made during the OHs time slot are not guaranteed to be answered during the same time slot.

Office Hours Etiquette Reminder

- Sign-up for a spot in the OH Queue: <https://ohq.eberly.cmu.edu/#/courses> and search for “14-763”
- Office Hours aim to help you find the path to maximize your learning experience.
- Getting the answers from the TA directly won't help you learn so **there won't be direct solutions provided during Office Hours.**
- The goal of the office hours is to **give you some ideas and pointers for you** to debug the issues.
- Please don't plan to spend **more than 10 minutes** in your conversation with the TA.
- Ask **good questions with due diligence.** Please research the issue and put an effort in implementing it before coming to Office hours.
 - **Example of a bad question:** I found this draft code online and I'm citing it but can't get it to work. Can you help?
 - **Example of a good question:** I'm getting a bug in my deployment to the cloud, I researched the issue and found these 3 different references (share the URLs). I implemented the first one and it didn't work. I'm trying the second one now and getting an error that I can't find enough references to it online. What could be the root cause of it?



Course Recitations

- Recitations offer important supplementary information, exercises or practices sets that are not covered in class.
- There are 10 recitations held throughout the semester. Refer to the course schedule for the corresponding weeks.
- Typically, recitations are held on Fridays of select weeks. Recitations will use the TA OH URL.
 - TAs may change the recitation date/time with a prior announcement on Piazza.
- Recitations are recorded and will be published on Canvas.
- While live attendance is not required, you may expect quiz questions covering notes presented in the recitations.



Recitation-1: System Setup

- This recitation aims to support the execution of part-1 of the homework.
- Students will receive supportive later this week to walk them through Part-1.
- TAs will hold additional recitations **this Friday** to respond to any inquiries/questions regarding the system setup based on your operating system.
 - Mac: Serena (2-3pm)
 - Linux: Tiffany (4-5pm)
 - Windows: Megha (6-7pm)

Course Assessment

Final Exam	Project	Assignments	Quizzes
25%	15%	40%	20%

- **Final Exam:** is an open-note test.
 - Students will have access to all the **PDFs** for lectures, readings and **HW** solutions. Students can bring any hard-copied materials with them.
 - Students are required to follow the schedule of their registered section. **On the scheduled final lecture of each section, final exam will be released only to the registered students of the corresponding section.** Each section will have its final exam version(s).
 - **Exam will be offered via [Lockdown Browser](#) and no knowledge exchange is allowed among students during the exam.**
 - Students are expected to install and test Lockdown browser on their machines ahead of the exam. If students face an issue with Lockdown browser installation, students must reach out to the instructors **no later than 2 weeks** before the final exam date.
 - **Sharing hard-copied notes is prohibited during the exam.**

Course Assessment – Cont'd

Final Exam	Project	Assignments	Quizzes
25%	15%	40%	20%

- **Course Project:** Each student will have the option to team up with another student for the project and you will choose one of two project options to submit. This project leverages most of the topics and practices that are covered throughout the semester. Course details are released in Week-3.
 - **Late submissions for the course project will receive no grade (0 points).**
- **Quizzes:** there will be 1 quiz published on Canvas during the lecture with a specific access code. The access code will be revealed during the lecture to the registered students of the corresponding section.
 - Quizzes will start from the third week of classes.
 - **Quizzes will cover lecture notes, readings, verbal explanations and recitations.**
 - If you encounter extenuating circumstances and must miss a class, please email the course instructor for taking a make-up quiz. Taking the make-up quiz is only available for legitimate extraordinary circumstances (e.g., illness, religious observance, conference attendance, job interviews).

Course Assessment – Cont'd

Final Exam	Project	Assignments	Quizzes
25%	15%	40%	20%

- **Homework Assignments:** there will be 8 homework assignments provided throughout the semester covering the practical aspects of the class. There will be good learning curve that students will have to take on their own.
- **Students will have 3 days to submit an assignment after the due date** and a late penalty will be applied. Late penalties are applied based on the timestamp of the last code commit on GitHub and it will follow this equation:
 - Total of 5 points for up to 24 hours delay
 - Total of 15 points for the next 24 hours delay
 - Total of 25 points for the next 24 hours delay
 - 100 points penalty (no grade) after this time.

After homework grades are released, **regrade requests can be made for 24 hours via Gradescope and CANNOT be submitted via email.**



Course Grading Scheme

+/- are used to provide granularity

Grade	Percentage Interval
A/A-	[85-100%], A starts from 93
B	[70-85%)
C	[55-70%)
D	[40-55%)
R (F)	Below 40%

Course Schedule

Week-1 (Aug 24)	- Introduction & Syllabus - System Setup - Dataset Introduction and Business Context - Introduction to the Cloud and Course Technology Stack	- System Setup HW released - System Setup Recitation	Mohamed
Week-2 (Aug 31)	- Introduction to Agentic Engineering - Data Collection and Storage • SQL and SparkSQL	- Introduction to Google Cloud & Agentic Engineering Recitation	Mohamed
Week-3 (Sep 7)	- Spark SQL and Data Frames	- System Setup HW deadline. - SparkSQL on PostgreSQL HW released. - Course Project Information Released - PySpark Recitation	Mohamed
Week-4 (Sep 14)	- NoSQL Database - Lab: Neo4j AuraDB	- SparkSQL on PostgreSQL HW deadline. - NoSQL & Kafka homework released - Neo4J Recitation	Mohamed (Sep 16 th lecture is pre-recorded).
Week-5 (Sep 21)	- Data Streaming - Lab: Confluent Kafka	- NoSQL & Kafka homework deadline. - Confluent Kafka Recitation	Mohamed
Week-6 (Sep 28)	- Data Engineering	- Course Project Checkpoint - Data engineering in Spark homework released - Data Engineering Recitation	Mohamed
Week-7 (Oct 5)	- SparkML– Training, Evaluation, Hyper-parameter Tuning - SparkML on the Cloud	- Data engineering in SparkML HW deadline - SparkML HW released - SparkML Recitation	Guannan

Course Schedule – Cont'd

Week-8 (Oct 19)	- Introduction to Pytorch - SGD & Neural Networks & Hyper-Parameter Tuning	- SparkML HW deadline	Guannan
Week-9 (Oct 26)	- Building LM with Pytorch - Computation Graph & GPU	- PyTorch HW released - PyTorch Recitation	Guannan
Week-10 (Nov 2)	- Distributed Training - Huggingface Ecosystem	- PyTorch HW deadline	Guannan
Week-11 (Nov 9)	- Advanced Topics in LLM - TensorFlow	- Course Project Deadline - LLM & TensorFlow & JAX HW release - LLM & TensorFlow Recitation	Guannan
Week-12 (Nov 16)	- JAX - TinyML	- TensorFlow & JAX HW deadline - TinyML HW released - Final Exam Recitation	Guannan/ Mohamed
Week-13 (Nov 23)	TinyML		Mohamed
Week-14 (Nov 30)	- ML Model Deployment to the Cloud & MLOps - Final Exam	- TinyML HW deadline	Mohamed

HW Submission Guidelines

- HW-1 will focus on Environment Setup and essential GitHub Skills.

- HW-1 submission:



- HW-2 and later assignment submissions:





Academic Integrity Violations (AIVs)

- AIVs are serious and can have direct impact on your course grade, your scholarship -if any-, your graduation timeline, and/or your continuation in your degree program.
- You may use all the materials and readings that we explicitly provide you with in the assignment document. In this case, attribution/citation is not required.
- You may use other material from the course, including lectures, Piazza posts by the instructors. For any such use, you must provide a proper attribution.
- You may not obtain code or other solution information from an unauthorized external source, including web pages, code repositories, blog posts, etc.
- You may not look at someone else's code (or other documents.) This includes one person looking at code and describing it to another. There is no notion of looking “too much,” since no looking is allowed at all.



Academic Integrity Violations (AIVs) – Cont'd

- For each assignment, the instructor may randomly invite a group of students to conduct oral discussions to learn how the students implemented the assignment. Students may face academic integrity violations if they fail to explain their own work. However, students can't use the direct homework question in their prompt to the AI tool.
- Students cannot publish their assignment solutions or project work publicly without an explicit approval from the instructor.
- Students are not allowed to share Quiz Access Code with their peers who didn't attend the class in person without pre-approval from the instructor. Sharing the quiz access code, without instructor's approval, is considered an academic integrity violation.



Our View on AI Tools

- AI can be a productive tool to your workflow. However, it can also be a diminisher to your critical skill abilities.
- In this course, our approach is **UBAS**.
- UBAS stands for: Understand the required concept(s), Build the necessary critical thinking abilities, Apply those abilities in a practical setting, Sstreamline your workflow, when appropriate, using AI tools.
- Students will have pre-identified tasks where they can use GenAI tools (including coding agents and chatbots). Generally, the use of GenAI tools is prohibited unless explicitly stated in the homework document. If explicitly stated, students can use GenAI tools and attribute the work generated by AI properly.

Why Not Overuse AI?

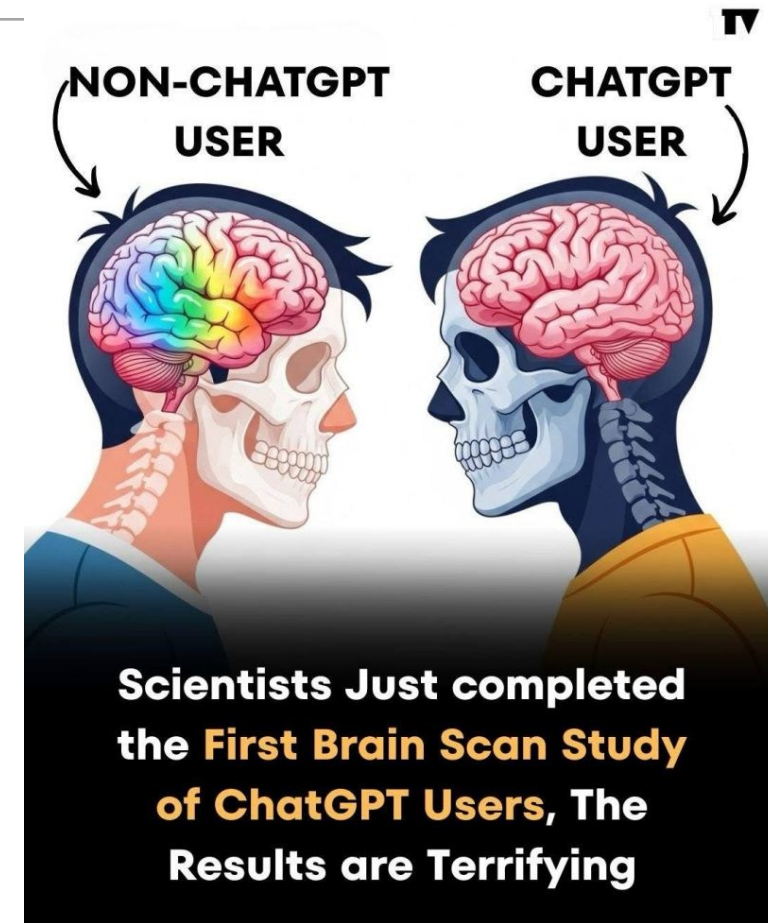
1. Concerns on the development of your basic cognitive skills.
2. Concerns on your creativity abilities.

AI in Writing Essays

From “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”:

<https://www.media.mit.edu/publications/your-brain-on-chatgpt/>

- Study compared three groups writing essays: LLM-assisted, Search Engine assisted, and Brain-only (no tools), over multiple sessions.
- Used EEG to measure brain connectivity; human and AI judges also scored essays.



Carnegie Mellon University

AI in Writing Essays – Findings

From “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”:

<https://www.media.mit.edu/publications/your-brain-on-chatgpt/>

- LLM assistance group suffered reduced neural connectivity, particularly weaker distributed networks, lower engagement relative to other groups.
- Over time, LLM users showed lower memory recall and more under-engagement; in swapped-to-brain-only condition, some reduced connectivity persisted.
- LLM group reported lowest sense of essay ownership; they had more difficulty accurately quoting their own writing.
- **Overusing LLM tools may bring cognitive costs, reduced mental effort, and engagement.**

AI in Ideation

From “The Ideation-Execution Gap: Execution Outcomes of LLM-Generated versus Human Research Ideas”: <https://arxiv.org/pdf/2506.20803>

- Generative AI generate research ideas that rate, initially, higher than human-expert ideas at ideation stage (novelty, excitement, expected effectiveness, overall quality).
- When these ideas are executed by expert implementers, the LLM ideas' scores drop much more than human ideas on all key metrics (novelty, excitement, effectiveness, overall quality).
- This gap (“ideation-execution gap”) is statistically significant: human ideas stay more stable; LLM ideas lose more of their initial promise.
- **Idea execution reveals factors often missed at ideations by LLMs such as empirical performance, rigorous experiment design, baseline comparisons, feasibility and resource constraints.**



Teaching Expectations

- Not all material is captured on the slides; students are expected to take notes during lectures.
- Students are expected to attend recitations (or watch the recordings) prior to the following lecture.
- Students are expected to complete all assigned readings and in-class exercises on-time.
- Instruction is delivered in a passionate teaching style. Don't be scared! 😊

Other Syllabus Information

- **If you run out of Google Cloud credits, plan for 24-48-hour delay to get a new coupon.**
- Syllabus contains important information about student wellness, student academic success center, and food insecurity.
- The Syllabus can be found on Canvas under the Modules section

Other Information

- **Your professional growth is of our utmost priorities.**
- **We will organize an NVIDIA workshop for fundamentals of accelerated computing during the weekend of the Fall break.**
- **If interested, fill the interest form on Canvas.**

Waitlisted?

For enrollment questions and inquiries, please email

1) INI Academics at
ini-academic@andrew.cmu.edu

OR

2) ECE Academics at
ece-academicaffairs@andrew.cmu.edu



Bottom Line

- You are a graduate student at CMU, and we expect you to **pay close attention to the details** mentioned during lectures and in your homework assignments.
- Ask questions and **avoid making unreasonable assumptions**.
- Use your **intellectual abilities and problem-solving skills** to fill in any gaps beyond what the teaching team can share with you.
- We are **committed to your success** both inside and outside the classroom.
- Feel free to **reach out with any questions** related to research, internships, career advice, or related topics.



Course Dataset



Course Dataset

- In this course, we will use the NSL-KDD dataset.
- The NSL-KDD dataset is an enhanced dataset to help researchers compare different intrusion detection methods.
- The dataset contains 100k+ records which is suitable for our class purposes
- You may download this dataset from Canvas Modules' section.



History of NSL-KDD Dataset

- Cybercrimes represent any criminal activity that involves a computer, a network, or a networked device. Cybercrimes may lead to physical damages or financial losses (in billions of dollars).
- It's critical to detect network intrusions before they occur. One way to identify intrusions is to look at previous potential intrusions and look for similarities/patterns. New intrusions are likely to share some aspects or features with previous intrusions. This field is called Intrusion Detection.
- In 1998, The **Defense Advanced Research Projects Agency (DARPA)** established the ****Intrusion Detection Evaluation Program**** to survey and evaluate research in intrusion detection. This program organized **The KDD cup** as an International Knowledge Discovery and Data Mining Tools Competition.

History of NSL-KDD Dataset – Cont'd

- In 1999, this competition was held with the goal of collecting traffic records. The competition task was to build a network intrusion detector, a predictive model capable of distinguishing between "bad" connections, called intrusions or attacks, and "good" normal connections. As a result of this competition, a mass amount of internet traffic records were collected and bundled into a data set called the **KDD'99**
- The **KDD'99 dataset** had several redundant records and issues with one column, so the **NSL-KDD dataset** was created as a newer version of it. We will use the NSL-KDD dataset in this course.



NSL-KDD Dataset – File Explanation

- **KDDTrain+.ARFF**: The full NSL-KDD train set with binary labels in ARFF format
- **KDDTrain+.TXT**: The full NSL-KDD train set including attack-type labels and difficulty level in CSV format
- **KDDTrain+_20Percent.ARFF**: A 20% subset of the KDDTrain+.arff file
- **KDDTrain+_20Percent.TXT**: A 20% subset of the KDDTrain+.txt file
- **KDDTest+.ARFF**: The full NSL-KDD test set with binary labels in ARFF format
- **KDDTest+.TXT**: The full NSL-KDD test set including attack-type labels and difficulty level in CSV format
- **KDDTest-21.ARFF**: A subset of the KDDTest+.arff file which does not include records with difficulty level of 21 out of 21
- **KDDTest-21.TXT**: A subset of the KDDTest+.txt file which does not include records with difficulty level of 21 out of 21

Try this at home!

Open the **KDDTrain text file** using Excel (or a Spreadsheet viewer) and validate these statistics

Dataset	Number of Records:					
	Total	Normal	DoS	Probe	U2R	R2L
KDDTrain+20%	25192	13449 (53%)	9234 (37%)	2289 (9.16%)	11 (0.04%)	209 (0.8%)
KDDTrain+	125973	67343 (53%)	45927 (37%)	11656 (9.11%)	52 (0.04%)	995 (0.85%)
KDDTest+	22544	9711 (43%)	7458 (33%)	2421 (11%)	200 (0.9%)	2654 (12.1%)



Next Steps

- Complete the Python survey if you haven't done so.
- Read “**A Detailed Analysis of the KDD CUP 99 Data Set.pdf**” published on Canvas.
- Prepare Jupyter Notebooks (or JupyterLab) to view future lectures and run code snippets.
- Join the course Piazza.
- Join the student Slack workspace.
- Familiarize yourself with the location of the in-person TA Ohs.

Waitlisted Students

